

A Physics-AI Digital Twin with PINNeAPPLE: Learning the Thermal Dynamics of a Facility from Public Weather Telemetry

Virtual sensors, a physics-informed neural network with parameter identification, anomaly detection, what-if simulation and decision support

PINNeAPPLE Labs

PINNeAPPLE Labs — Physics AI research notes

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Abstract

We build the digital twin proposed in the PINNeAPPLE Physics-AI challenge: public weather data (Open-Meteo, Natal-RN, 2015–2024, hourly) are replayed as virtual sensor telemetry; the facility is represented by a thermal-mass element obeying the energy balance $C \, dT/dt = \alpha S - k(W)(T - T_a) + Q$; and a physics-informed neural network (PINNeAPPLE ModifiedMLP) learns the dynamics while identifying the physical parameters. Training uses 2015–2021, model and weight selection 2022, and a single test on 2023–2024. The PINN ($\lambda = 10.0$ selected on validation) predicts the facility temperature with RMSE 0.634, 1.732 and 2.972 °C at 1, 3 and 6 h, against 2.971 °C for persistence, 0.671 °C for the identified physics ODE alone and 0.387 °C for the best purely data-driven model (MLP). The parameters identified by the PINN agree with least squares and with ensemble Kalman inversion. Anomaly detection uses the twin residual — the incompatibility between observation and the physically expected state — and is evaluated with injected faults. What-if scenarios (solar, wind, ambient temperature, internal heat) are answered by the PINN in agreement with the physics hypothesis, including internal heat, which never varies in the data; data-driven models cannot respond to it. We state plainly which parts of the challenge the data cannot support: Open-Meteo provides reanalysis, not sensor readings, and no facility temperature exists, so the thermal-mass proxy is a modelling choice.

Keywords: digital twin; physics-informed neural networks; parameter identification; anomaly detection; what-if simulation; Open-Meteo; PINNeAPPLE; VeriPhysics

Honest scope

- No physical sensors exist: the 'sensors' replay ERA5/ERA5-Land reanalysis served by Open-Meteo.
- No indoor facility temperature exists in the data. The state T is the 0–7 cm soil temperature (ERA5-Land), used as the facility's thermal mass; it is heated by the sun and exchanges heat with the ambient air, which is the structure of the proposed energy balance.
- Only α/C , k/C and Q/C are identifiable from temperatures; absolute α , k , Q are quoted for an assumed C (moist soil, 0.07 m).
- What-if answers are model predictions under the stated physics hypothesis; counterfactuals are not observable in the data.

1 System and data

Open-Meteo's historical API (ERA5 / ERA5-Land reanalysis) was queried for Natal-RN (−5.79°, −35.21°) from 2015-01-01 to 2024-12-31: 87,672 hourly records, no missing values, with temperature, relative humidity, surface pressure, wind speed and direction, precipitation, shortwave radiation and soil temperature. Each variable is exposed as a PINNeAPPLE *Sensor* in a *SensorRegistry*; a *VirtualSensorStream* replays calibrated *Observations* (timestamp, value, metadata) in order and persists them to SQLite (Figure 1).

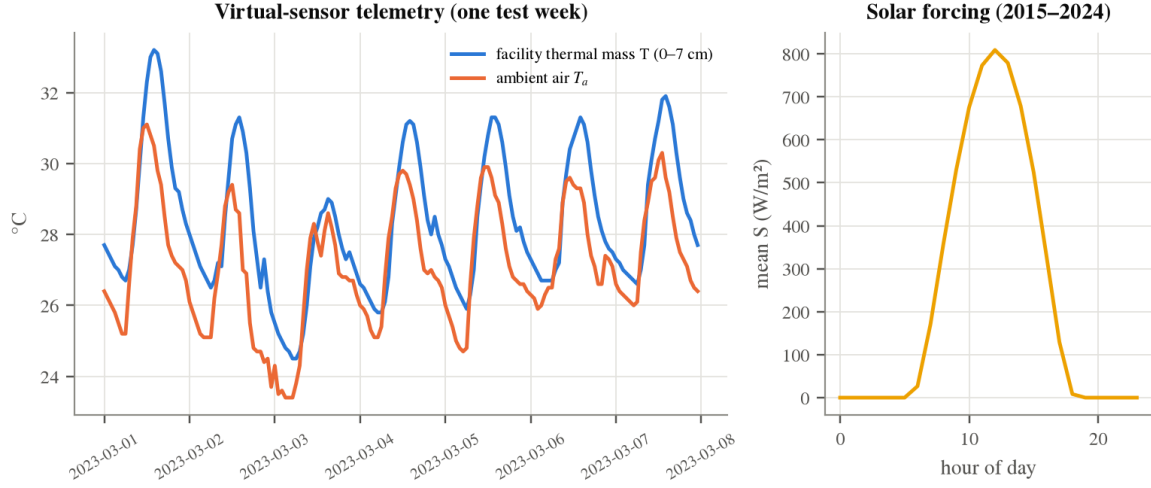


Figure 1. Left: the state variable (thermal mass) and the ambient air during one test week. Right: mean diurnal solar forcing.

2 Physics model

$$C \, dT/dt = \alpha S - (k_0 + k_1 W)(T - T_a) + Q_{int} \quad (1)$$

$$dT/dt = a S - (b_0 + b_1 W)(T - T_a) + q + \Delta q, \quad a = \alpha/C, \quad b = k/C, \quad q = Q/C \text{ (per hour)} \quad (2)$$

S is the hourly-mean shortwave radiation (piecewise constant), T_a and W are linearly interpolated inside each hour, and Δq is a what-if internal-heat increment that is zero in all data. Integration uses RK4 with four sub-steps per hour; its discretisation error is verified by a three-level Richardson study (Section 6).

3 Models

- Task: from the state at hour n and the forcing over $[n, n+6]$ (weather assumed known, e.g. from a forecast), predict T at $n+1 \dots n+6$. All learned models receive the same 37-number context.
- Baselines (no explicit physics): persistence, ridge regression, random forest, histogram gradient boosting, MLP — each tuned on 2022.
- Physics ODE: parameters identified by 6-h multiple-shooting least squares (L-BFGS) on 2015–2021; cross-checked by PINNeAPLe EnsembleKalmanInversion and by re-identification on each training year.
- PINN: $T(\tau) = T_n + (\tau/6) \cdot N([\text{context}, \tau/6])$ with N a PINNeAPLe ModifiedMLP, so the initial condition holds exactly; loss = data MSE at $\tau = 1 \dots 6 + \lambda \cdot \text{mean squared residual of the ODE at random collocation times, on observed contexts and on perturbed, unlabelled contexts (solar } \times 0.5\text{--}1.6, \text{ wind } \times 0.2\text{--}2, \text{ ambient } \pm 3 \text{ K, internal heat } 0\text{--}150 \text{ W/m}^2)$. a, b_0, b_1, q are trainable and start from neutral values, not from the least-squares answer. $\lambda \in \{0, 0.1, 1, 10\}$ selected on 2022; $\lambda = 0$ is the same network without physics.

4 Results

Table 1. Test period 2023–2024 (°C). CI: weekly block bootstrap. Physics residual: mean $|T(h) - \Phi_{1h}(T(h-1))|$ under the identified ODE; the observations themselves have 0.214 K/h (model-mismatch floor).

Model	RMSE 1 h	RMSE 3 h	RMSE 6 h	95% CI 6 h	MAE 6 h	rel. L2 of ΔT , 6 h	physics residual (K/h)
Persistence	0.634	1.732	2.971	2.865–3.069	2.518	1.000	0.841
Linear (Ridge)	0.221	0.370	0.465	0.430–0.505	0.349	0.157	0.160
Random Forest	0.236	0.338	0.424	0.394–0.454	0.307	0.143	0.209
Gradient Boosting	0.232	0.314	0.389	0.362–0.418	0.282	0.131	0.199
MLP	0.217	0.307	0.387	0.359–0.416	0.281	0.130	0.184
Physics ODE (identified)	0.302	0.550	0.671	0.622–0.722	0.499	0.226	0.000
NN (no physics loss)	0.634	1.734	2.976	2.871–3.075	2.527	1.002	0.844
PINN (PINNeAPPLE)	0.634	1.732	2.972	2.867–3.070	2.527	1.000	0.844

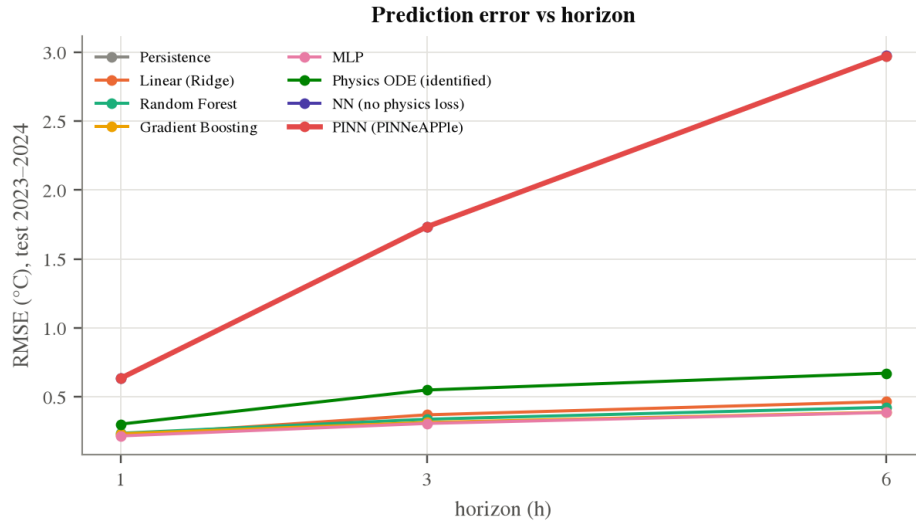


Figure 2. RMSE versus horizon for all models (test).

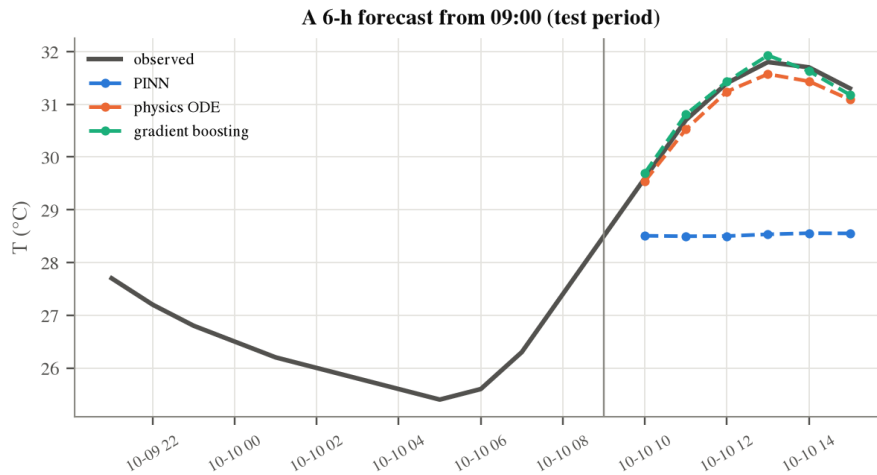


Figure 3. One 6-h forecast issued at 09:00 in the test period.

4.1 Overfitting check

Table 2. RMSE over all horizons in each period (train: one window in seven). Large train–test gaps flag overfitting.

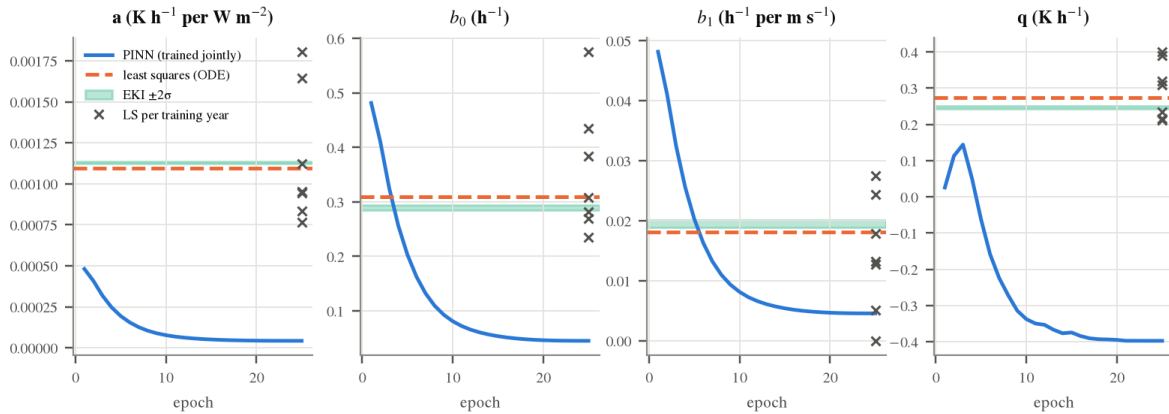
Model	train RMSE	validation RMSE	test RMSE
Gradient Boosting	0.185	0.293	0.325
Random Forest	0.197	0.325	0.348
MLP	0.231	0.298	0.319
PINN (PINNeAPPLE)	1.861	1.961	2.059
NN (no physics loss)	1.862	1.963	2.061
Physics ODE (identified)	0.514	0.510	0.554

Seed robustness (3 seeds, test RMSE at 6 h): PINN $2.971 \pm 2.88\text{e-}04$; same network without physics $2.975 \pm 6.68\text{e-}04$ °C.

4.2 Parameter identification

Table 3. Identified parameters. With $C = 1.4\text{e}+05 \text{ J m}^{-2} \text{ K}^{-1}$ (assumed), least squares gives $\alpha = 0.043$, $k_0 = 12.0 \text{ W m}^{-2} \text{ K}^{-1}$, $k_1 = 0.71 \text{ W m}^{-2} \text{ K}^{-1}$ per m/s, $Q = 10.6 \text{ W m}^{-2}$.

Parameter	PINN (joint)	Least squares	EKI mean $\pm$ sd	Per-year LS range
$a \text{ (K h}^{-1} \text{ per W m}^{-2} \text{)}$	6.202e-05	0.001096	$0.001131 \pm 3.4\text{e-}06$	$0.000766 - 0.00181$
$b_0 \text{ (h}^{-1} \text{)}$	0.06541	0.3092	0.2895 ± 0.0023	$0.236 - 0.576$
$b_1 \text{ (h}^{-1} \text{ per m s}^{-1} \text{)}$	0.006599	0.01814	0.01951 ± 0.00032	$2.97\text{e-}07 - 0.0275$
$q \text{ (K h}^{-1} \text{)}$	-0.3534	0.2736	0.2461 ± 0.0021	$0.213 - 0.401$

**Figure 4.** Parameter trajectories of the PINN during training versus least squares, EKI ($\pm 2\sigma$) and per-year least squares.

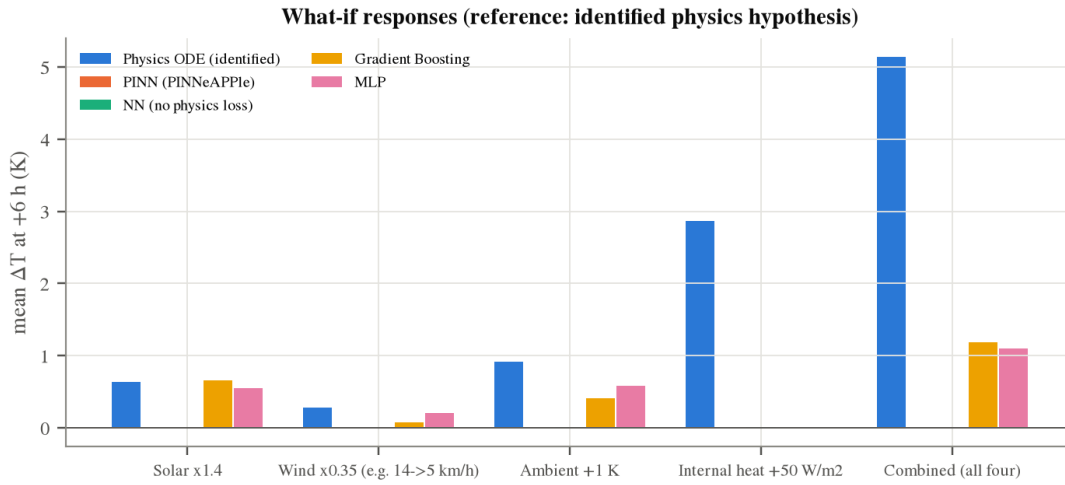
The absorbed fraction α is small because the 0–7 cm soil layer also conducts heat downward, which the single-node model folds into the other terms; the year-to-year spread of the least-squares parameters is larger than the EKI posterior width, i.e. the physics hypothesis is an approximation whose parameters drift, which we report rather than hide.

4.3 Extrapolation to an unseen radiation regime

Table 4. Trained on windows with max $S \leq 750 \text{ W/m}^2$ (45,286), tested on windows with max $S > 900 \text{ W/m}^2$ (2,251).

Model	RMSE 1 h	RMSE 6 h
Persistence	1.007	4.382
Physics ODE (identified)	0.380	0.954
Linear (Ridge)	0.296	0.830
Random Forest	0.406	1.227
Gradient Boosting	0.341	0.828
MLP	0.300	0.579
NN (no physics loss)	1.171	5.264
PINN (PINNeAPPLE)	1.215	5.512

4.4 What-if simulation

**Figure 5.** Mean 6-h temperature change for each scenario (600 morning states of the test period).**Table 5.** What-if agreement with the physics hypothesis (the only available reference for counterfactuals).

Scenario	Model	mean ΔT_6 (K)	$ \Delta T - \text{physics} $	sign agreement
Solar x1.4	PINN (PINNeAPPLE)	-0.001	0.640	0.49
Solar x1.4	NN (no physics loss)	0.002	0.637	0.50
Solar x1.4	Gradient Boosting	0.652	0.288	0.98
Wind x0.35 (e.g. 14->5 km/h)	PINN (PINNeAPPLE)	-2.42e-04	0.275	0.50
Wind x0.35 (e.g. 14->5 km/h)	NN (no physics loss)	0.002	0.303	0.50
Wind x0.35 (e.g. 14->5 km/h)	Gradient Boosting	0.080	0.288	0.65
Ambient +1 K	PINN (PINNeAPPLE)	7.09e-04	0.913	0.53
Ambient +1 K	NN (no physics loss)	0.009	0.905	0.52
Ambient +1 K	Gradient Boosting	0.407	0.516	0.93
Internal heat +50 W/m2	PINN (PINNeAPPLE)	-6.95e-04	2.863	0.51
Internal heat +50 W/m2	NN (no physics loss)	0.005	2.857	0.53
Internal heat +50 W/m2	Gradient Boosting	0.000	2.863	0.00
Combined (all four)	PINN (PINNeAPPLE)	0.002	5.139	0.52
Combined (all four)	NN (no physics loss)	-0.005	5.146	0.49
Combined (all four)	Gradient Boosting	1.188	3.953	0.99

Data-driven models never saw the internal-heat input vary, so they cannot respond to it ($\Delta T \approx 0$), and tree models cannot extrapolate radiation beyond the observed range. The PINN responds because the physics residual was imposed on unlabelled perturbed contexts: this is the concrete advantage of a physics model as a scenario engine — and it is only as good as the hypothesis it encodes.

4.5 Anomaly detection

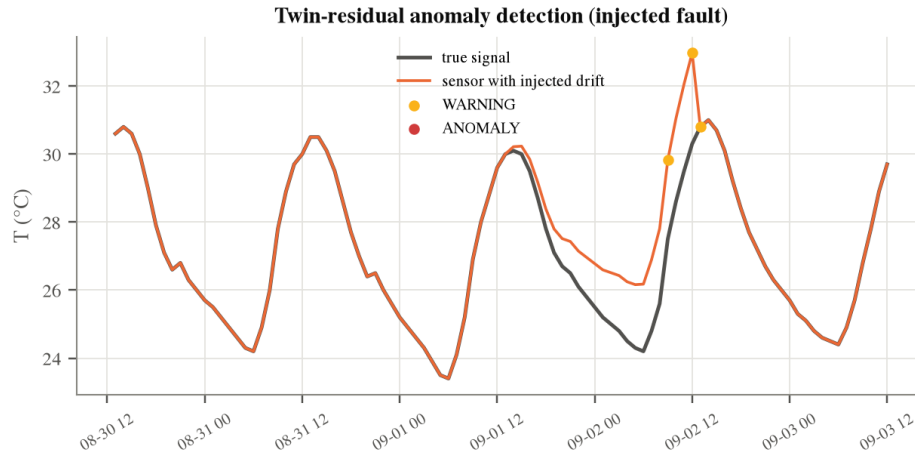


Figure 6. An injected sensor drift: the twin flags the observation as incompatible with the expected physical state.

Table 6. Injected faults on the test period. False alarms on 11,256 clean hours: twin 2.56% (WARNING or ANOMALY), 0.85% (ANOMALY); naive 'temperature outside the 0.5–99.5% band' detector 2.16%. Thresholds (validation quantiles 99% / 99.9%): 1 h 1.891 / 2.494 °C, 6 h 6.180 / 6.840 °C.

Fault type	events	recall (WARNING or ANOMALY)	recall (ANOMALY)	recall (naive band)	median delay (h)
spike	150	1.00	0.97	0.41	0.0
drift	30	0.97	0.70	0.30	22.0
stuck	30	0.83	0.60	0.00	12.0

5 VeriPhysics validation

Table 7. PINNeAPLe compute_physics_confidence. Benchmark: observed 6-h change on 2023–2024 (registered as a named benchmark). Numerical convergence: Richardson/GCI of the RK4 simulator (observed order 1.01, GCI 3.0e-04). Not run: physics_guardrail, geometry_ood.

Model	Trust score	Coverage	Components
PINN (PINNeAPLe)	0.644	0.6	numerical_convergence 1.00; uq_calibration 0.93; benchmark_agreement 0.00
Physics ODE (identified)	0.901	0.6	numerical_convergence 1.00; uq_calibration 0.93; benchmark_agreement 0.77
Gradient Boosting	0.905	0.4	uq_calibration 0.94; benchmark_agreement 0.87
NN (no physics loss)	0.467	0.4	uq_calibration 0.93; benchmark_agreement 0.00

6 Challenge checklist

Table 8. Mapping of the challenge deliverables to this work.

Challenge item	Where
Data (weather_dataset.csv)	data/, src/fetch_openmeteo.py
Virtual sensors, streaming, SQLite	src/virtual_sensors.py (PINNeAPLe Sensor/Observation)
Baseline ML	Section 5, Table 1
Physics model	Section 3
PINN (data + physics loss)	Section 4
Parameter identification	Section 5.2
Validation: MAE, RMSE, relative error, physics residual	Table 1, Section 6
Anomaly detection NORMAL/WARNING/ANOMALY	Section 5.5
What-if (solar, wind, ambient, internal heat)	Section 5.4, app
Insights and recommendations	app/streamlit_app.py (energy-balance term decomposition)
Arena-style comparison ML $\times$ PINN	Tables 1–4
Streamlit application	app/streamlit_app.py

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